

LLM Based Mental Trait Evaluation From Online Interviews

Students:

Hanah Shlomo and Malkiel Gal

M.Sc. Project

Faculty of Data Science, HIT- Holon Institute of Technology

Supervisor:

Dr. Gaddi Blumrosen

In collaboration with Dr. Hila Gvirts, Ariel University

Scope

- Scientific Background
- Motivation
- Methods
- Results
- Discussion

Background

Mental traits

We focused on three mental traits:

Alexithymia

PTSD

ADHD

What is Alexithymia

- A personality trait characterized by difficulty identifying and describing one's feelings
 - ✓ DIF -difficulty identifying feelings
 - ✓ DDF - difficulty describing feelings
- Associated with an externally oriented thinking style (EOT)
 - ✓ Focusing on concrete external details rather than internal experiences

“no words for emotion”



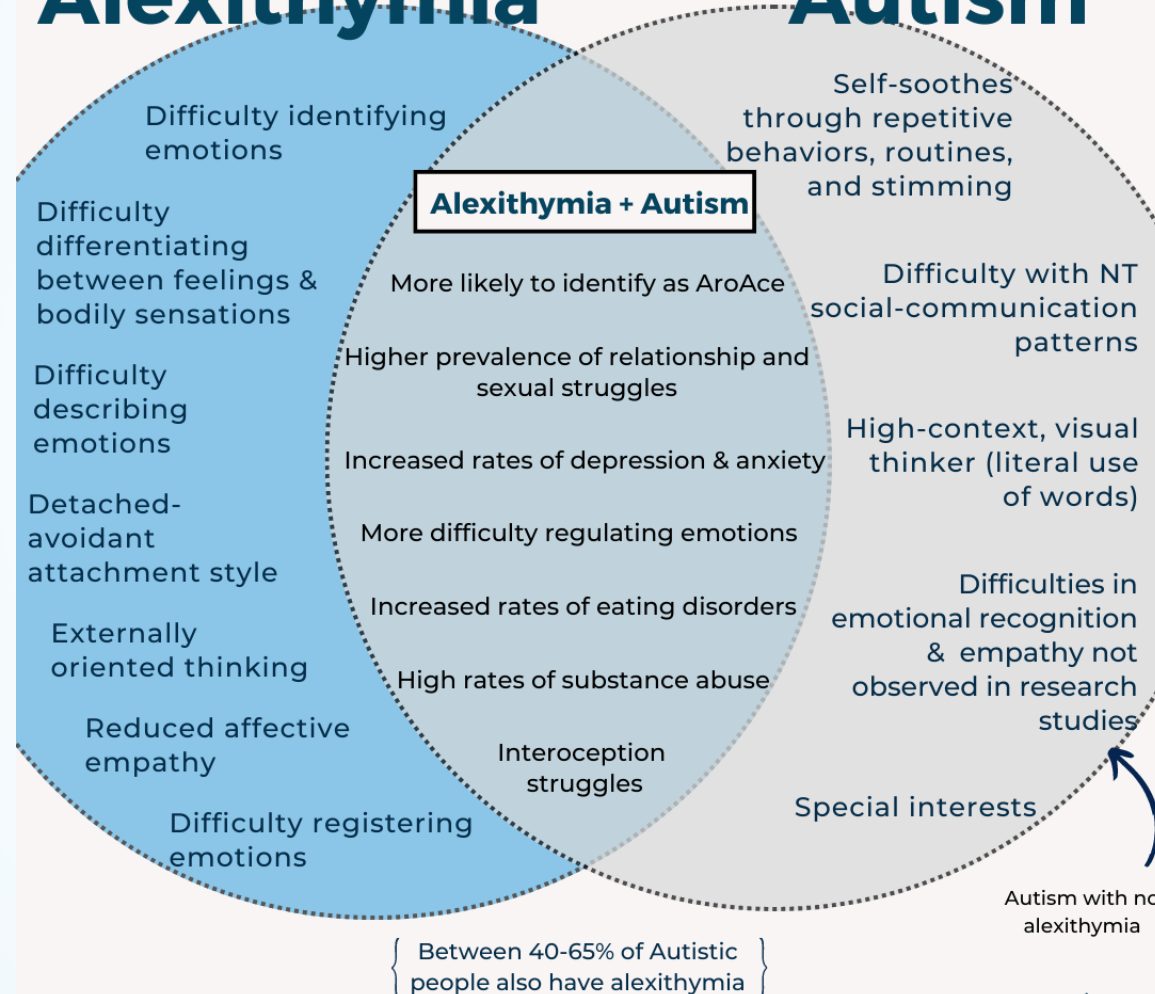


Misdiagnosis Monday

Alexithymia and Autism

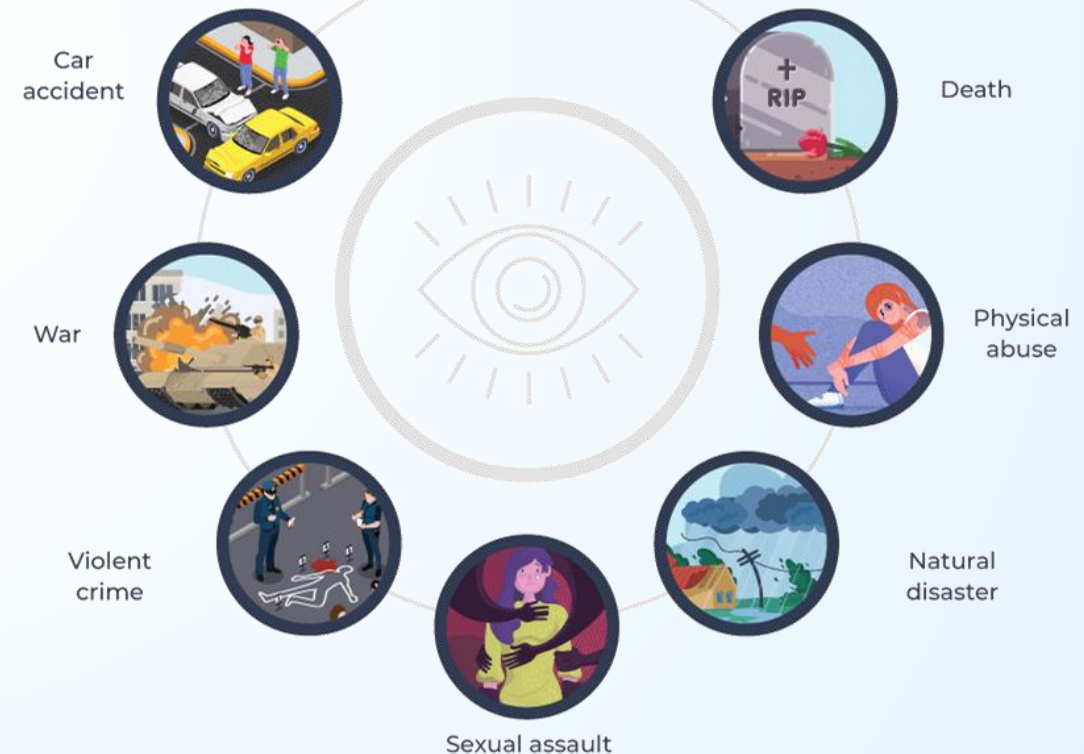
Alexithymia

Autism



What is **Post-traumatic stress disorder (PTSD)**?

- Post-Traumatic Stress Disorder (PTSD) is a mental health condition
- Triggered by
 - ✓ Terrifying, shocking, scary, or dangerous event
 - ✓ Experience or witness



PTSD symptoms

Post-Traumatic Stress Disorder (PTSD)

- Begin within 3 months of the traumatic event
- Chronic PTSD
 - ✓ After 3 months
 - ✓ Affect life functioning



Avoid Thinking of the Trauma



Avoid Talking of the Trauma



Easily Frightened



Negative Mood



Negative Thinking



Always on Guard



Avoiding Places



Avoiding Activities



Flashbacks



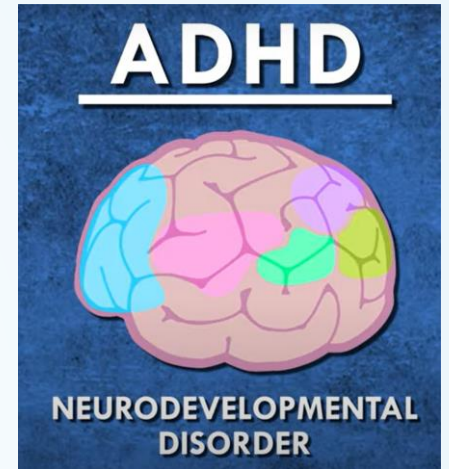
Cannot Concentrate



Aggressive Behavior

What is ADHD

- Attention Deficit (or) Hyperactivity Disorder (ADHD)
- ADHD is the most common behavioral and neurological disorder
- ~ 20% diagnosed children with ADHD
- Around 40% on the spectrum of ADHD



ADHD symptoms

Inattentive



Easily distracted



Unorganized



Difficulty listening

Hyperactive/Impulsive



Difficulty sitting still



Rush through tasks



Make rash decisions

Combined



Symptoms of
both presentations

Motivation

Need

Accurate assessment for treatment
optimal plan design

Early treatment can ease symptoms and
save life

Current evaluation methods

- Mostly clinical
- Aided by self report questionnaires



Alexithymia self report

1. Total Severity analysis
 2. Separate **Alexithymia** biomarker (symptom) severity
 3. Separate Questionary answers (sub-score) severity
- The TAS-20 has 3 subscales:
 - Difficulty Describing Feelings subscale: 5 items – 2, 4, 11, 12, 17.
 - Difficulty Identifying Feeling: 7 items – 1, 3, 6, 7, 9, 13, 14.
 - Externally-Oriented Thinking : 8 items – 5, 8, 10, 15, 16, 18, 19, 20.

TAS-20

		Strongly disagree	Disagree	Neutral	Agree	Strongly agree
1	I am often confused about what emotion I am feeling.	1	2	3	4	5
2	It is difficult for me to find the right words for my feelings.	1	2	3	4	5
3	I have physical sensations that even doctors don't understand.	1	2	3	4	5
4	I am able to describe my feelings easily.	5	4	3	2	1
5	I prefer to analyze problems rather than just describe them.	5	4	3	2	1
6	When I am upset, I don't know if I am sad, frightened, or angry.	1	2	3	4	5
7	I am often puzzled by sensations in my body.	1	2	3	4	5
8	I prefer to just let things happen rather than to understand why they turned out that way.	1	2	3	4	5
9	I have feelings that I can't quite identify.	1	2	3	4	5
10	Being in touch with emotions is essential.	5	4	3	2	1
11	I find it hard to describe how I feel about people.	1	2	3	4	5
12	People tell me to describe my feelings more.	1	2	3	4	5
13	I don't know what's going on inside me.	1	2	3	4	5
14	I often don't know why I am angry.	1	2	3	4	5
15	I prefer talking to people about their daily activities rather than their feelings.	1	2	3	4	5
16	I prefer to watch "light" entertainment shows rather than psychological dramas.	1	2	3	4	5
17	It is difficult for me to reveal my innermost feelings, even to close friends.	1	2	3	4	5
18	I can feel close to someone, even in moments of silence.	5	4	3	2	1
19	I find examination of my feelings useful in solving personal problems.	5	4	3	2	1
20	Looking for hidden meanings in movies or plays distracts from my enjoyment.	1	2	3	4	5

PTSD evaluation scale

- 22-item questionnaire (self report)
- Part of DSM-5 for PTSD
- Self-report rating scale is 0-4

Item	Subscale	Content
Intrusion		
1	I1	Any reminder brought back feelings about it.
2	I2	I had trouble staying asleep.
3	I3	Other things kept making me think about it.
6	I4	I thought about it when I didn't mean to.
9	I5	Pictures about it popped into my mind.
14	I6	I found myself acting or feeling ...
16	I7	I had waves of strong feelings about it.
20	I8	I had dreams about it.
Avoidance		
5	A1	I avoided letting myself get upset ...
7	A2	I felt as if it hadn't happened or wasn't real.
8	A3	I stayed away from reminders of it.
11	A4	I tried not to think about it.
12	A5	I was aware that I still had a lot of feelings ...
13	A6	My feelings about it were kind of numb.
17	A7	I tried to remove it from my memory.
22	A8	I tried not to talk about it.
Hyperarousal		
4	H1	I felt irritable and angry.
10	H2	I was jumpy and easily startled.
15	H3	I had trouble falling asleep.
18	H4	I had trouble concentrating.
19	H5	Reminders of it caused me to have physical ...
21	H6	I felt watchful and on-guard.

ADHD evaluation scale

Adult adhd self-report scale (asrs-v1.1) symptom checklist part of DSM-5 for **ADHD**

- Items reflects
 - ✓ Changes to existing symptoms
 - ✓ Addition of new symptoms in DSM-5
- Self-report rating scale is 1-5

ADULT ADHD SELF-REPORT SCALE (ASRS-V1.1) SYMPTOM CHECKLIST

Patient: _____ Date Completed: _____

Please answer the questions below, rating yourself on each of the criteria shown using the scale on the right side of the page. As you answer each question, place an X in the box that best describes how you have felt and conducted yourself over the past 6 months. Please give this completed checklist to your healthcare professional to discuss during your appointment.	Never	Rarely	Sometimes	Often	Very often
PART A					
How often do you have trouble wrapping up the final details of a project, once the challenging parts have been done?					
How often do you have difficulty getting things in order when you have to do a task that requires organization?					
How often do you have problems remembering appointments or obligations?					
When you have a task that requires a lot of thought, how often do you avoid or delay getting started?					
How often do you fidget or squirm with your hands or feet when you have to sit down for a long time?					
How often do you feel overly active and compelled to do things, like you were driven by a motor?					
PART B					
How often do you make careless mistakes when you have to work on a boring or difficult project?					
How often do you have difficulty keeping your attention when you are doing boring or repetitive work?					
How often do you have difficulty concentrating on what people say to you, even when they are speaking to you directly?					
How often do you misplace or have difficulty finding things at home or at work?					
How often are you distracted by activity or noise around you?					
How often do you leave your seat in meetings or in other situations in which you are expected to stay seated?					
How often do you feel restless or fidgety?					
How often do you have difficulty unwinding and relaxing when you have time to yourself?					
How often do you find yourself talking too much when you are in social situations?					
When you're in a conversation, how often do you find yourself finishing the sentences of the people you are talking to, before they can finish it themselves?					
How often do you have difficulty waiting your turn in situations when turn taking is required?					
How often do you interrupt others when they are busy?					

Current diagnostics approach limitations

Clinical evaluation Limitations

- Patient Travel time
- Mental Health time
- Bias in assessment

Questionary (self-assessment) Limitations

- Reliability and Validity
- Children limitations

Research goal

→ Need for automated approach for evaluation

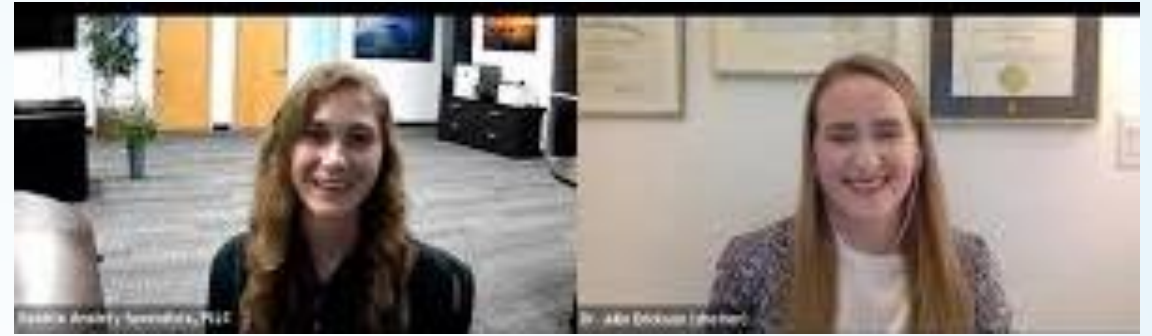
We suggest using the real-life text from
accessible media

Beyond Self-Reports: Unveiling mental traits with AI

- To overcome existing assessment limitations, we propose leveraging AI-based tools, specifically Large Language Model (LLM) for mental trait recognition and evaluation
- LLM can recognize patterns within the text by utilizing its Artificial Intelligence
- LLM common utilities:
- ChatGPT (OpenAI), Gemini (Google), Claude (Anthropic)

Methods

Experimental setup



Subjects:

20 Individuals exposed to Trauma



Online interviews via Zoom (Video) were recorded



Reference Labels:

Participants filled out self report questionnaires: TAS-20, IES-R, ASRS

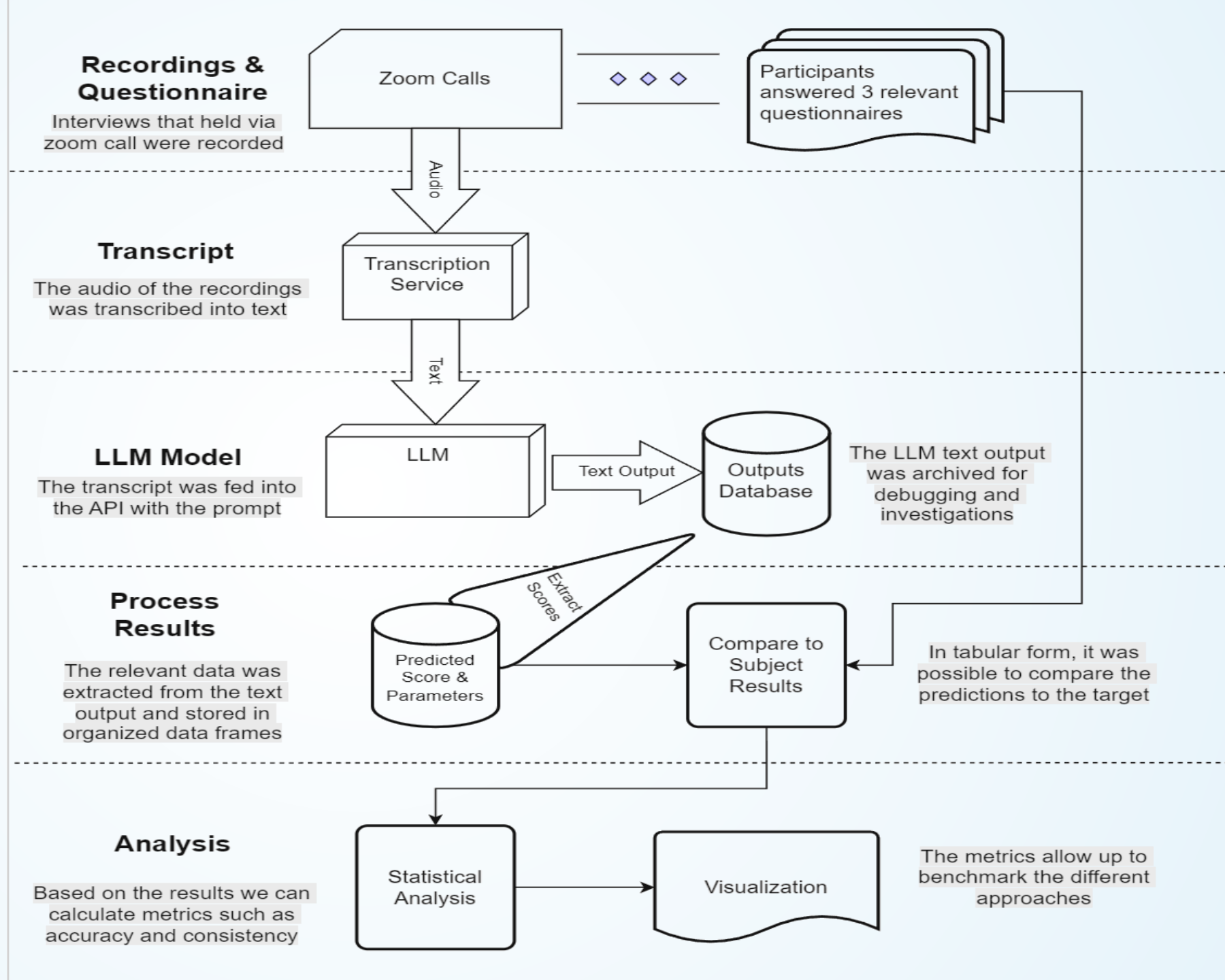


Audio was transcribed to text with Amazon-Transcribe and speaker recognition feature.

Method

1. Conduct Self report for the mental traits (Questionnaires)
2. Record interview
 - Subject of interest with mental health professional
3. Voice to text transcribe
4. Deploy Large Language Models (LLMs) to evaluate the mental trait severity
 - Automated prompts
5. Compare the self report to the LLM based estimation
6. Find real-life biomarkers and insights

Algorithm Workflow



Prompt Design

Design Criterion	Implementation
Low Computational Resources	Pre-trained LLMs Output tuned by Prompt Engineering
Efficient testing	Modular structure
Accuracy	Assessment compared real results
Consistency	Multiple iterations for each prompt
Reliability	Compare between different prompts

LLM & Prompt Design

We tested different approaches:

- **Direct** - Evaluate the total score directly, without references.
- **Sub-score** - Evaluate symptom scores, without references.
- **Items** - Predict the questionnaire score, with references.
 - Predict each question (item) in the questionnaire.

Prompt Components

Each improvements to the naïve approach we call **Prompt Component**:

Prompt Component	Details
Instructions	General instructions, Assume a role of a mental health expert.
Context	Background information, key terms, assessment goals, and diagnostic criteria.
Chain of thoughts	Step-by-step instructions, structured reasoning process.
Questionnaire	Text copy of the relevant questionnaire (TAS-20, IES-R, or ASRS)
Example Interview Example Evaluation	A sample interview, expected output format.
Interview	The transcript of the interview for assessment.
Disclaimer	A statement clarifying the usage of the assessment <i>“This assessment is intended as a decision support tool and is not a substitute for a clinical diagnosis by a qualified mental health professional.”</i>

Permutation Parameters Structure

Parameter	Details
Prompt Type	Direct, Sub-score, Items
Condition	The mental condition or disorder that is evaluated. (Alexithymia, PTSD, ADHD)
Questionnaire	TAS-20, IES-R, ASRS
Q Format	The format of the questionnaire. (Compact, Detailed)
Instructions	The variant of the instruction
Context	The variant of the context.
Chain of Thoughts	The variant of the detailed instructions.
Example Interview	The variant of the provided transcript example. Includes the speaker 1 only variation as well
Example Evaluation	The variant of the evaluation example. Conditions the LLM to output the answer in a certain format
Mental State	Part of the example evaluation component. Whether the answer should include a list of Mental State Indicators
Prediction Output Format	CSV, Natural Language
Disclaimer	Whether to include a disclaimer at the end of the prompt
Batch	A grouping index for the predefined permutations
Unique Index	A unique enumerator for the attempted iteration

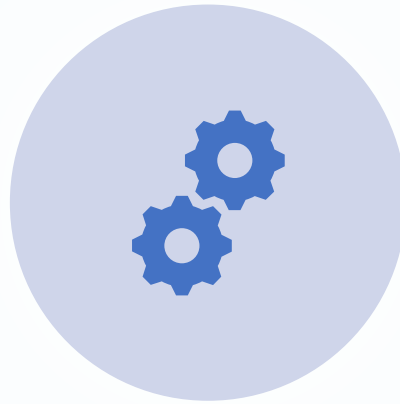
Scores adjustment

- Linear transformation is necessary to adjust the LLM's output.
- Let $T: A \rightarrow B$ be a linear transform between scale A to scale B .
- $$\begin{cases} T: [A_{min}, A_{max}] \rightarrow [B_{min}, B_{max}] \\ T(x) = B_{min} + (x - A_{min}) \cdot \frac{(B_{max} - B_{min})}{(A_{max} - A_{min})} \end{cases}$$
- For example :
- $$\begin{cases} T_{alexithymia}: [1, 10] \rightarrow [20, 100] \\ T_{alexithymia}(score) = 20 + \frac{(score - 1) \cdot (100 - 20)}{(10 - 1)} \end{cases}$$

Performance metrics



ACCURACY



CONSISTENCY
(RELIABILITY)

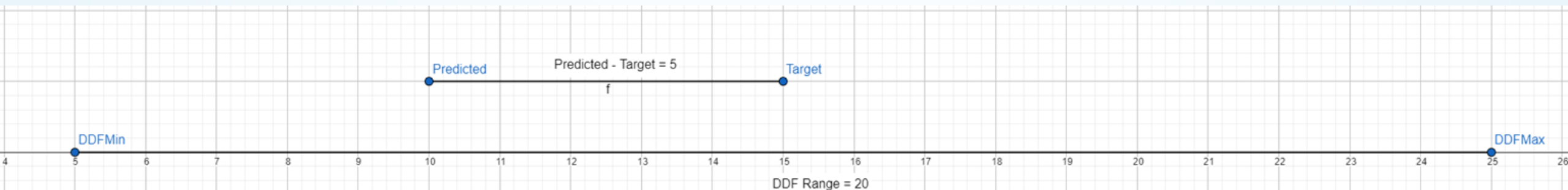


CONFIDENCE

Accuracy

- Error is the relative distance between the predicted score and real score, out of the whole subscore range. Error: $\epsilon(y) = \left| \frac{Y - y'}{B_{max} - B_{min}} \right|$
- Accuracy is the complementary part to the error. $Accuracy = 1 - \epsilon$
- For example - Let's take distance of 5 over a range of 20 (like in DDF)

Real Score	Predicted	Error	Accuracy
10	15	(15-10)/20 25%	1-0.25 75%



Consistency (Reliability)

- We calculated the Consistency errors as the coefficient of variation
- *Coefficient of Variation* $= \frac{\sigma}{\mu}$
- *Mean* (μ): $\mu = \frac{(\Sigma x)}{n}$
- *Standard Deviation* (σ): $\sigma = \sqrt{\left[\frac{(\Sigma (x - \mu)^2)}{n-1} \right]}$

Confidence

- Confidence of the answer reflects how much the LLM is sure in his answer in scale 0-1.
- Indication for bad interview



Results and Discussion

Permutation Guide

- Direct approach - General Score:

Attempting to evaluate the final condition score without references. With Explanation and Confidence Level

- **[1]** General Score - Patient Only
 - **[2]** General Score - Both Speakers
-

- Direct approach - Per Subscore:

Attempting to evaluate each subscore of the condition separately, without references, with Explanation and Confidence Level

- **[3]** Subscores - Patient Only
 - **[4]** Subscores - Both Speakers
-

- Based on questionnaires - Per Item:

Attempting to predict the condition score according to the commonly used self-administered questionnaire for that condition. In order to "stick to the source", we predict each question (item) in the questionnaire.

- **[5]** Per Item - Minimum Components:

- Context
- Chain of Thoughts
- Questionnaire

- **[6]** Per Item - Some Components

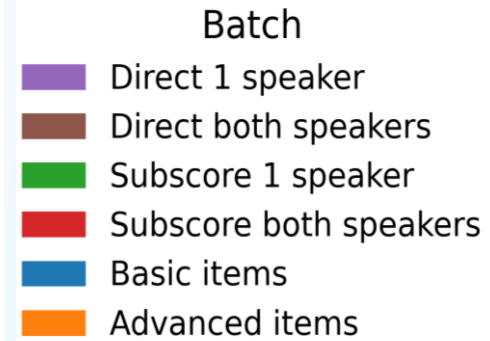
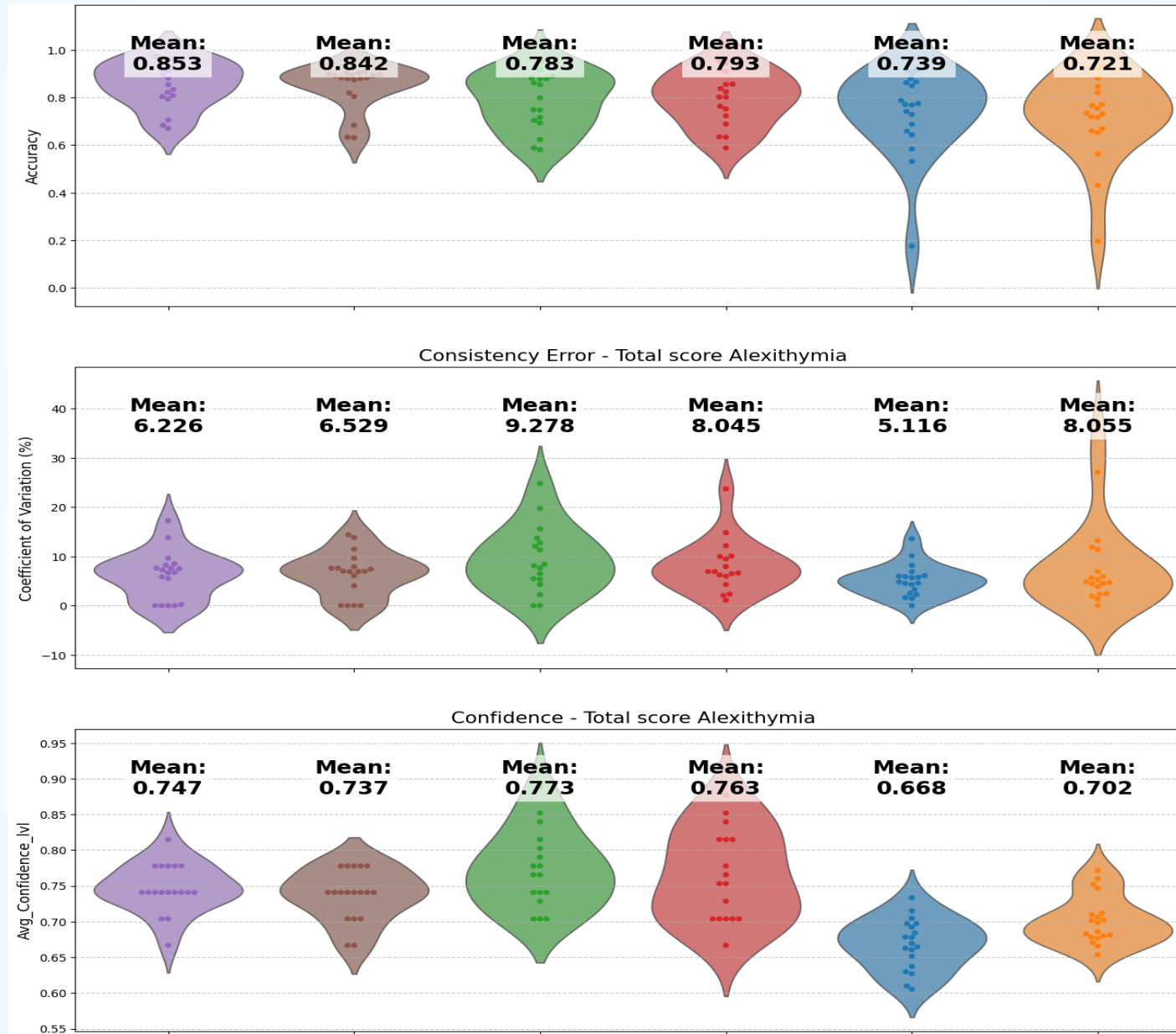
- Instructions
- Context (predict score)
- Chain of Thoughts (subscores only)
- Questionnaire
- Example Evaluation (csv only)

- **[7]** Per Item - Maximum Components

- Instructions
- Context (threefold)
- Chain of Thoughts (step by step)
- Questionnaire
- Example Interview
- Example Evaluation (with Mental State Indicators)
- Disclaimer

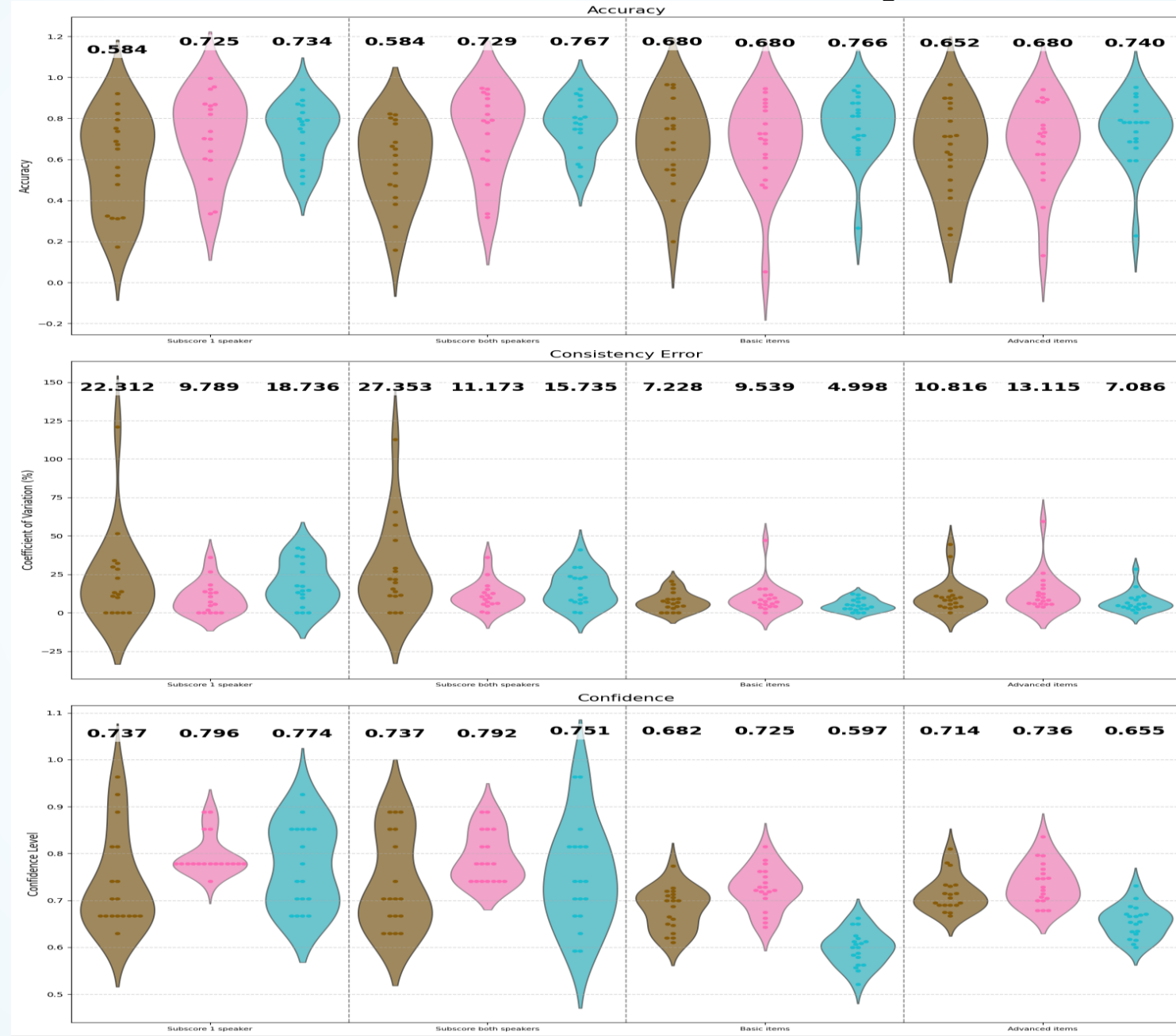
Performance Metrics - Alexithymia

- Direct approach achieved the best accuracy with 85.3%
- All prompts were consistent
- Sub score approach achieved the best confidence with 77.3%



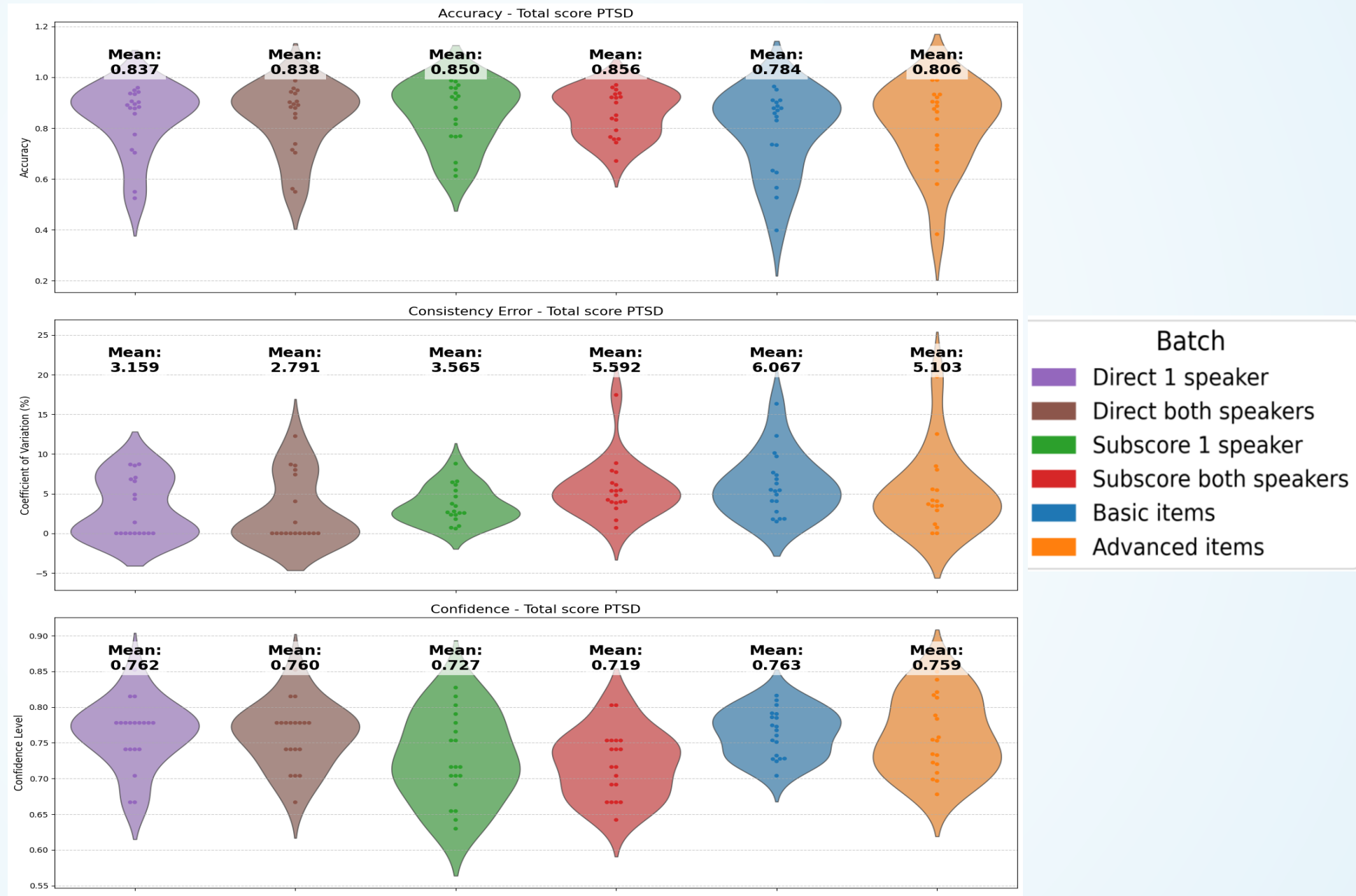
Performance Metrics - Alexithymia

- EOT achieved the highest accuracy across all sub scales around 75%.
- Lower accuracy compared to total score prompt
- High consistency error
- Overall same confidence level compared to total score prompt



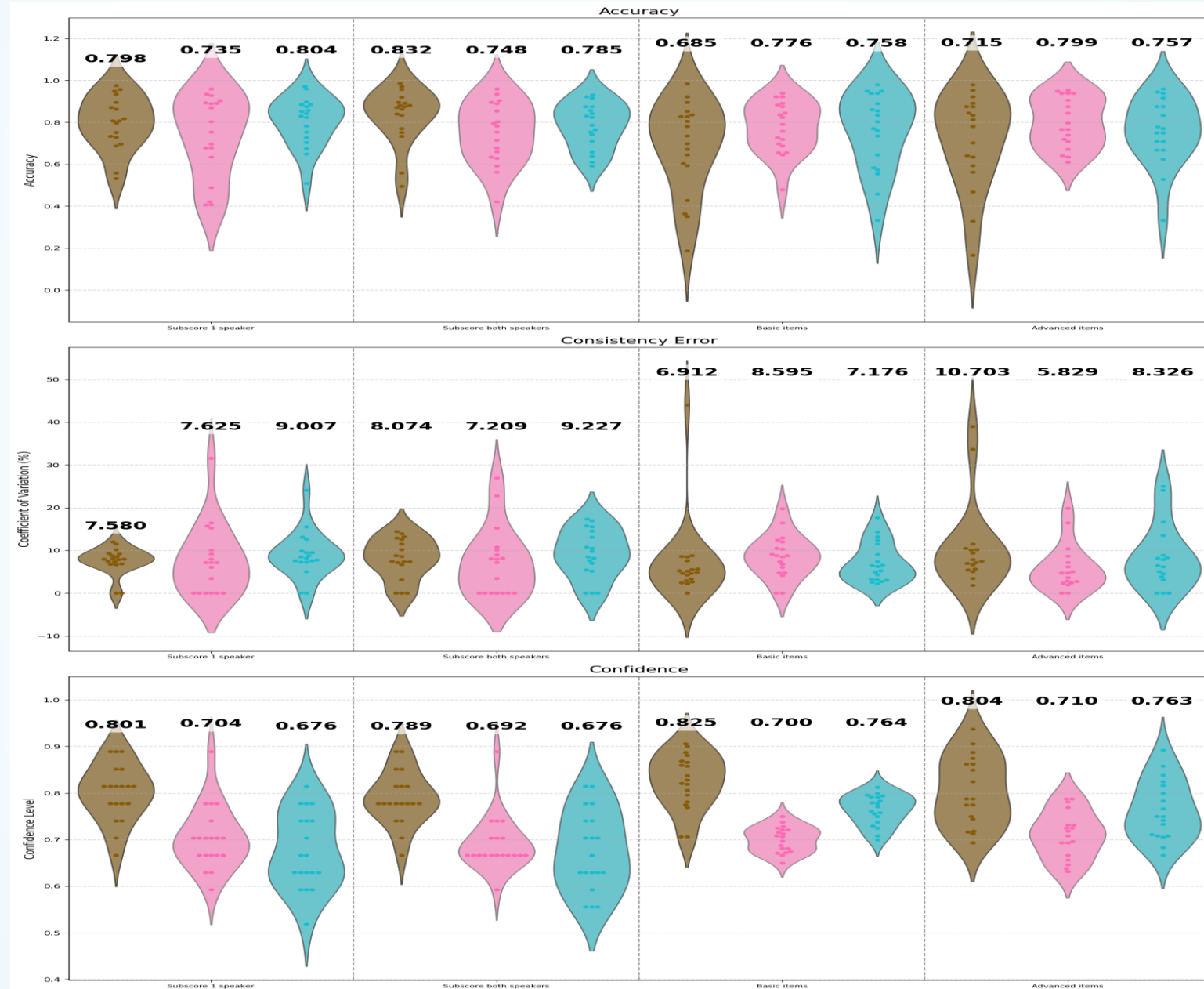
Performance Metrics - PTSD

- Sub score approach achieved the best accuracy with 85%
- All prompts were very consistent
- Overall confidence around 75%



Performance Metrics - PTSD

- Hard to decide which approach is the best
- Lower accuracy compared to total score prompt
- Overall same confidence level compared to total score prompt



Subscore

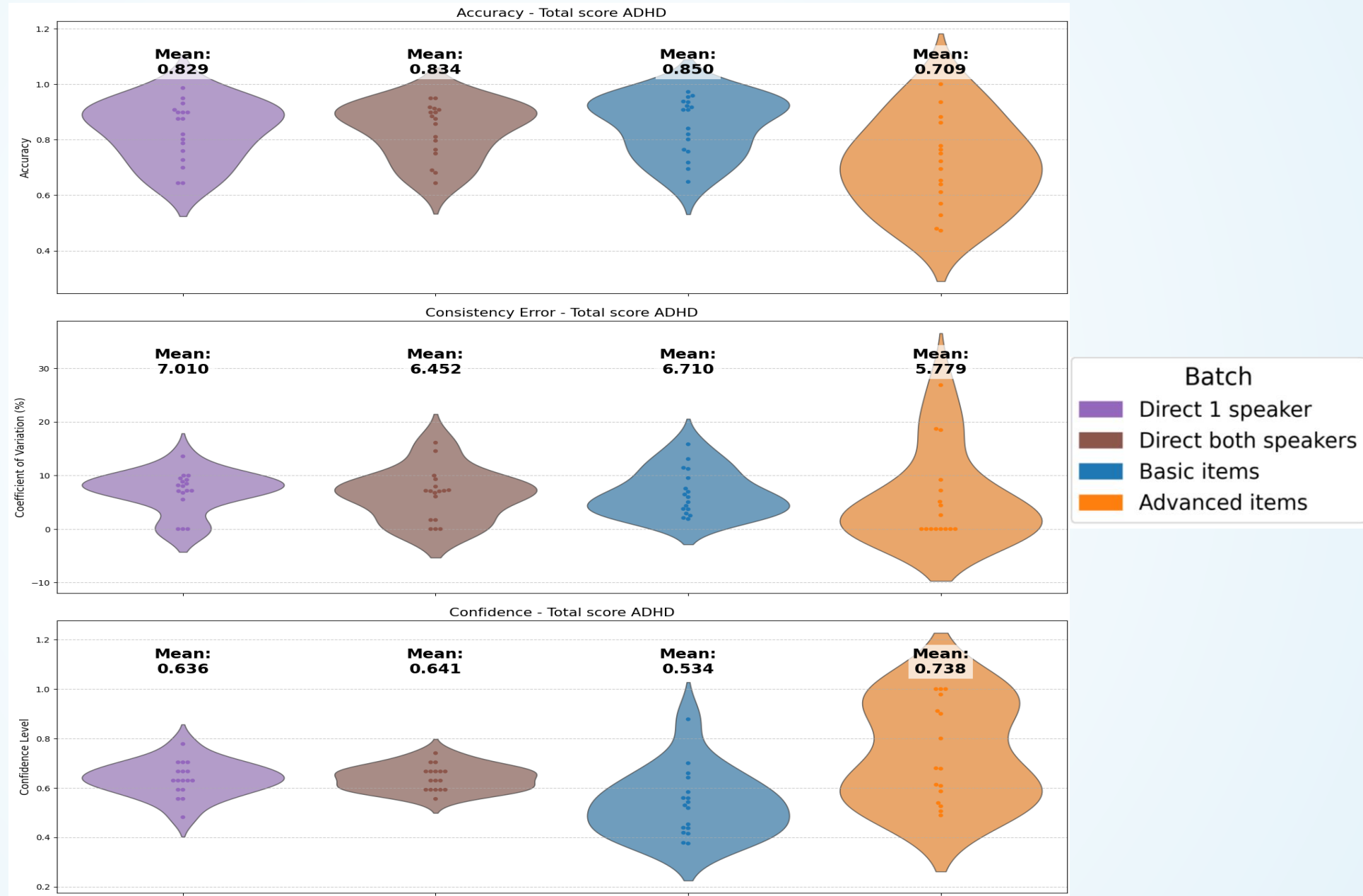
Intrusion

Avoidance

Hyperarousal

Performance Metrics - ADHD

- Basic items approach achieved the best accuracy with 85%
- All prompts were very consistent
- Lowest confidence achieved the highest accuracy

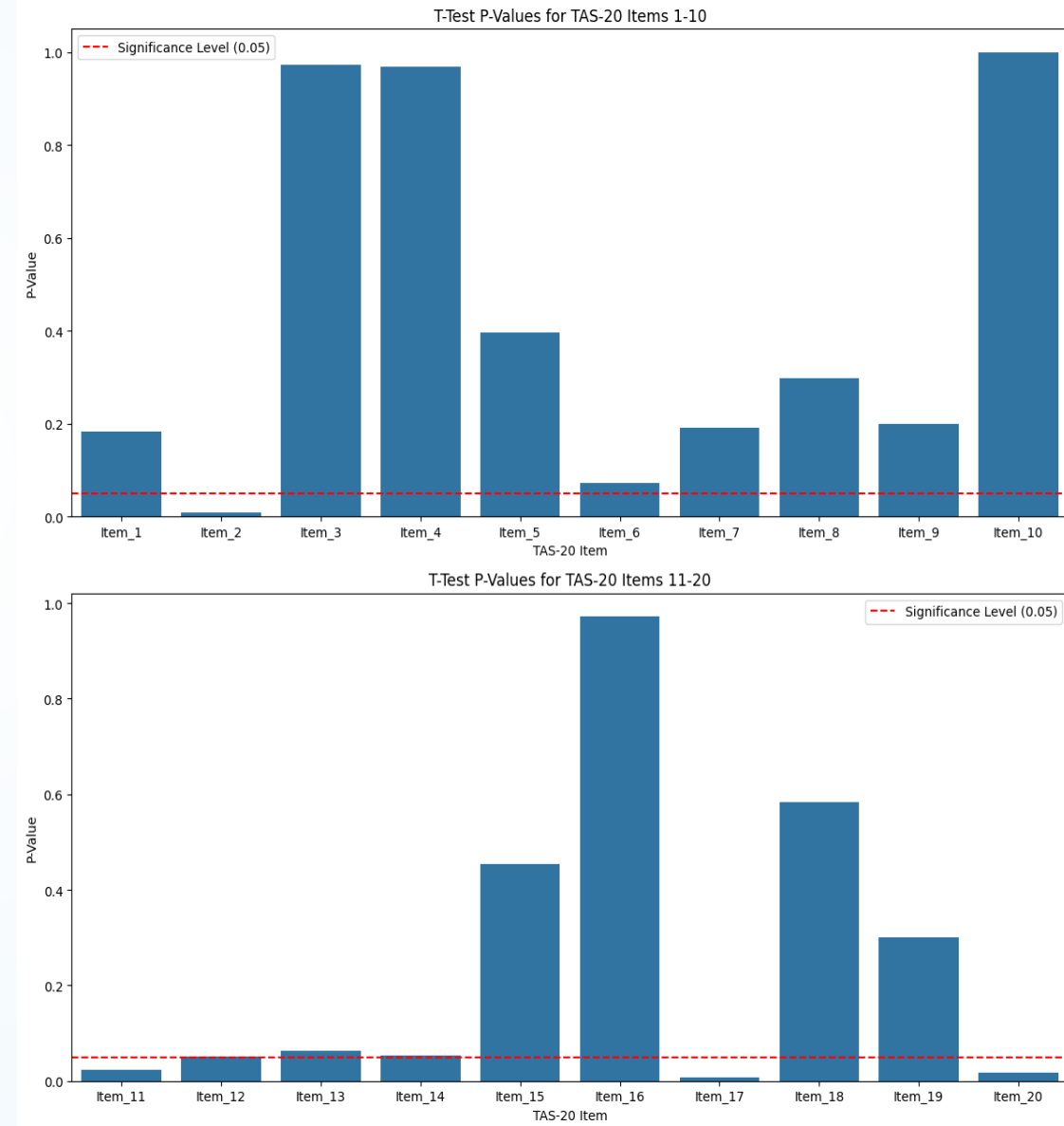


Statistical analysis

1. T test
 - ✓ **Goal:** Identify significant differences between groups
 - ✓ **Method:** Compare means of variables
2. Chi square
 - ✓ **Goal:** Assess normality of data distribution
 - ✓ **Method:** Tested for goodness of fit
3. Power analysis
 - ✓ **Purpose:** Determine required sample sizes
 - ✓ **Method:** Calculate sample sizes

Statistics tests – T test Alexithymia

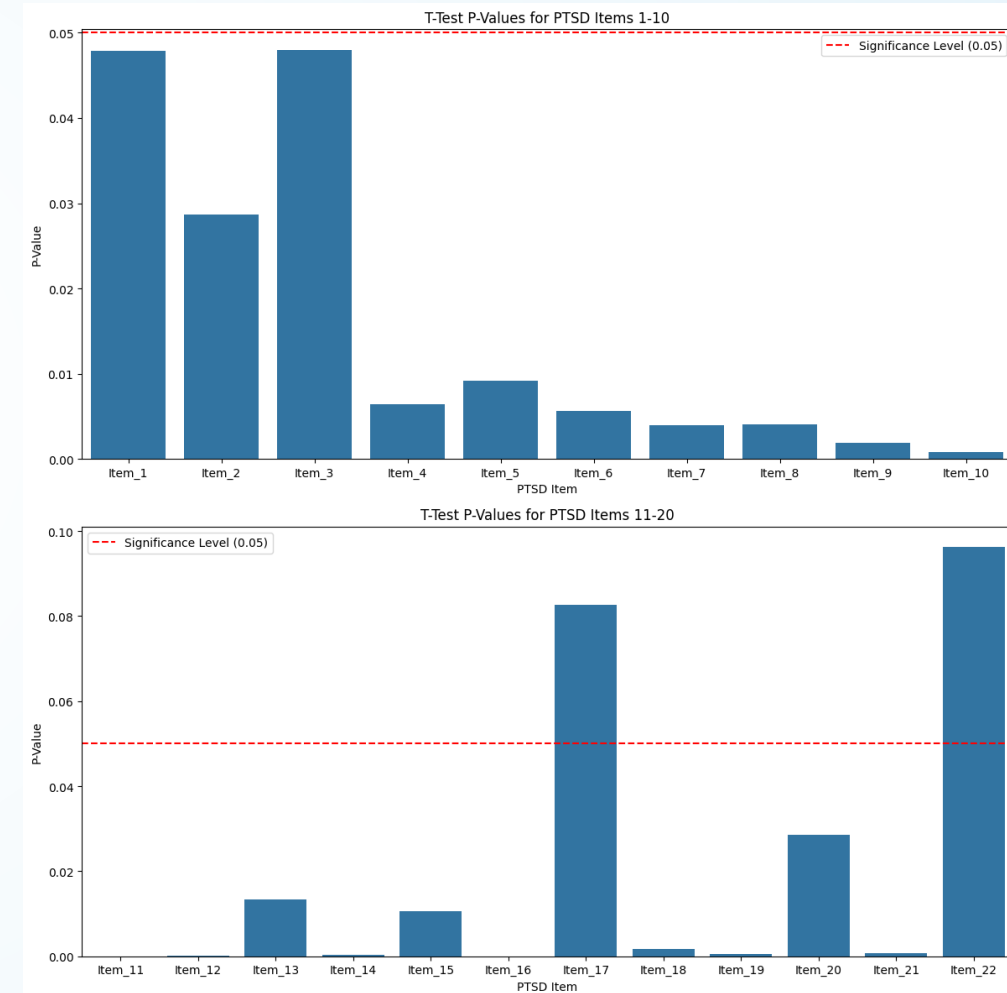
- Separated into 2 group by score of 51
 - DDF - items 2, 11, 12, and 17.
 - EOT – item 20.
-
- ❑ 12 – “People tell me to describe my feelings more”
 - ❑ 20 – “I look for hidden meanings in movies or plays”



Statistics tests – T test PTSD

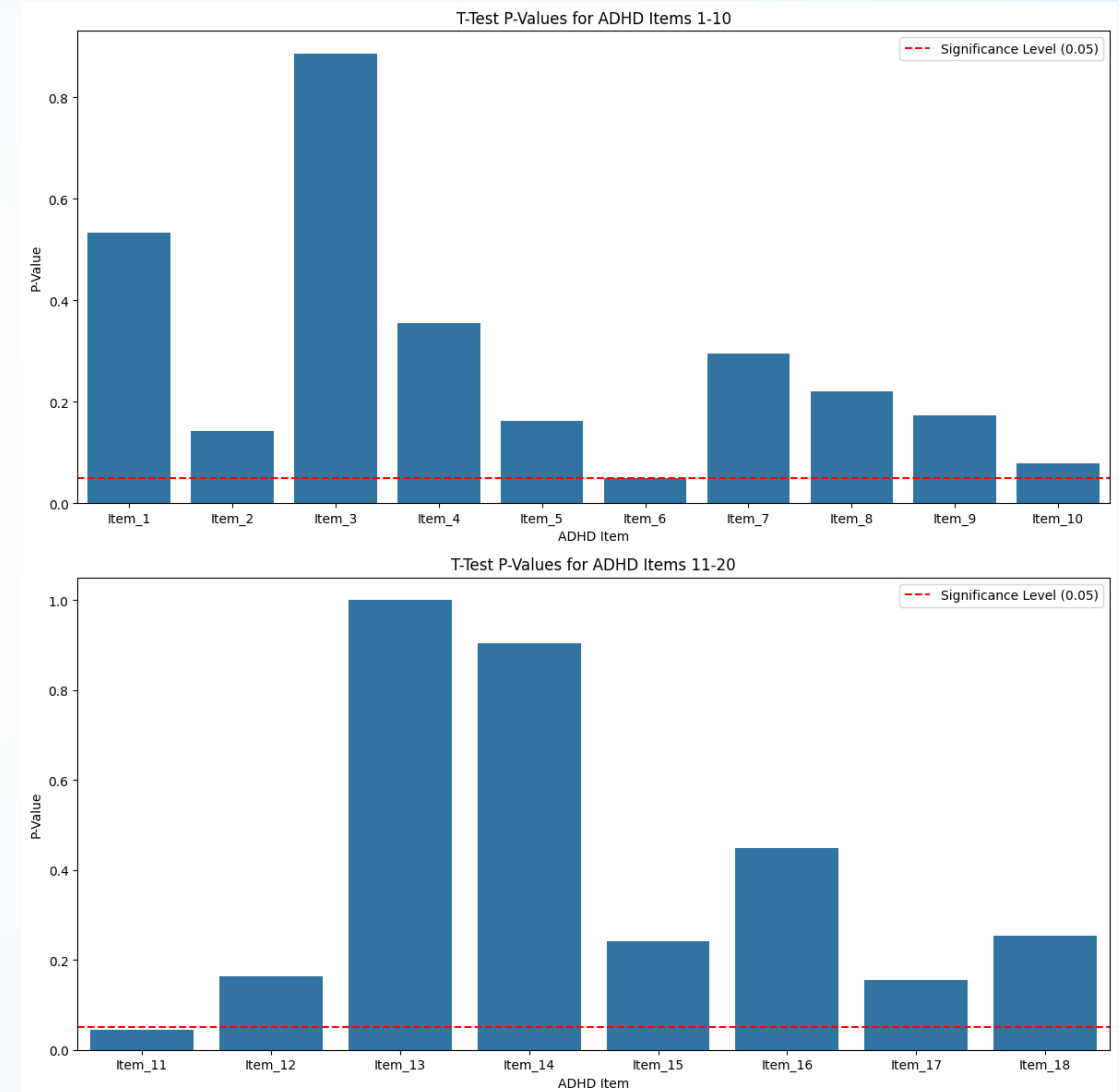
- Separated into 2 group by score of 37
- All items except **17** and **22**.
- Most items are highly sensitive
- Complexity of avoidance
- Biased subjective answers

- ☐ 17 – “I tried to remove it from my memory”
- ☐ 22 – “I tried not to talk about it”



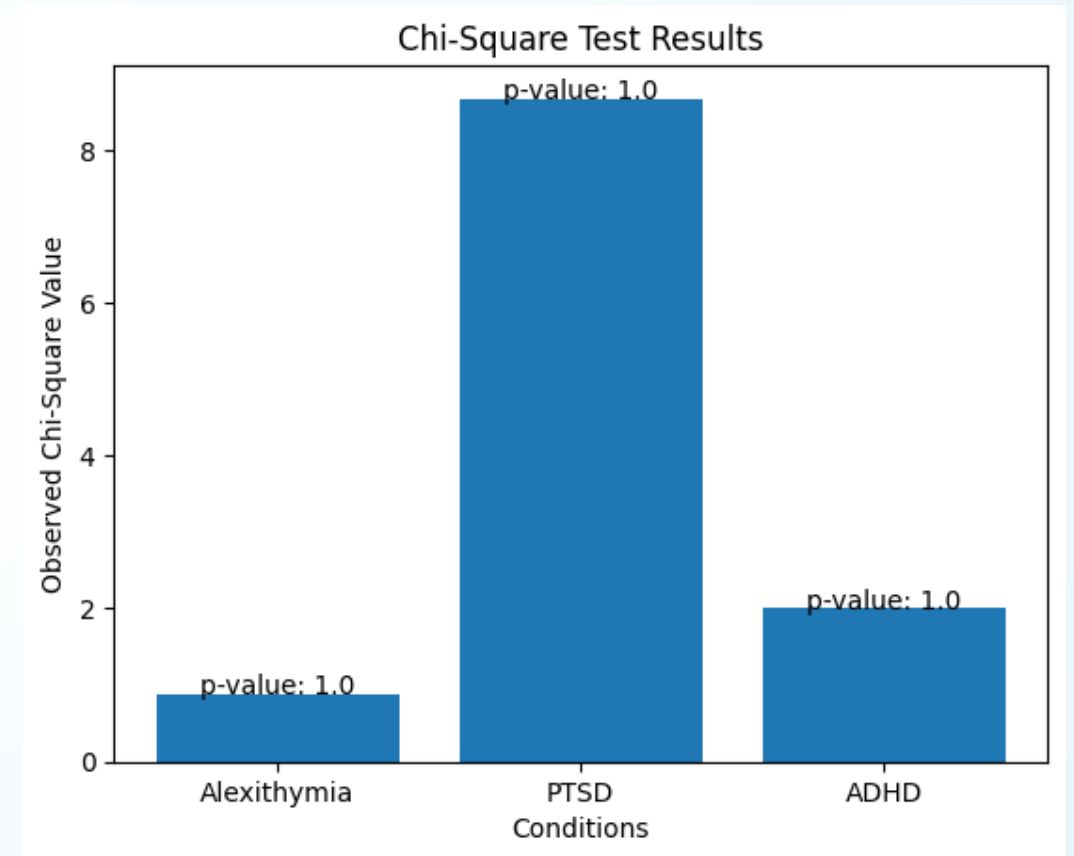
Statistics tests – T test ADHD

- Separated into 2 group by score of 43
- Items 6 and 11.
- **Limitations** in capturing the full spectrum of ADHD symptoms
 - ❑ 6 – “How often do you feel overly active and compelled to do things, like you were driven by a motor”
 - ❑ 11 – “How often are you distracted by activity or noise around you”



Statistics tests - Chi Square

- **Normal Distribution:** The data distributions for all three conditions (Alexithymia, PTSD, and ADHD) do not significantly deviate from a normal distribution.

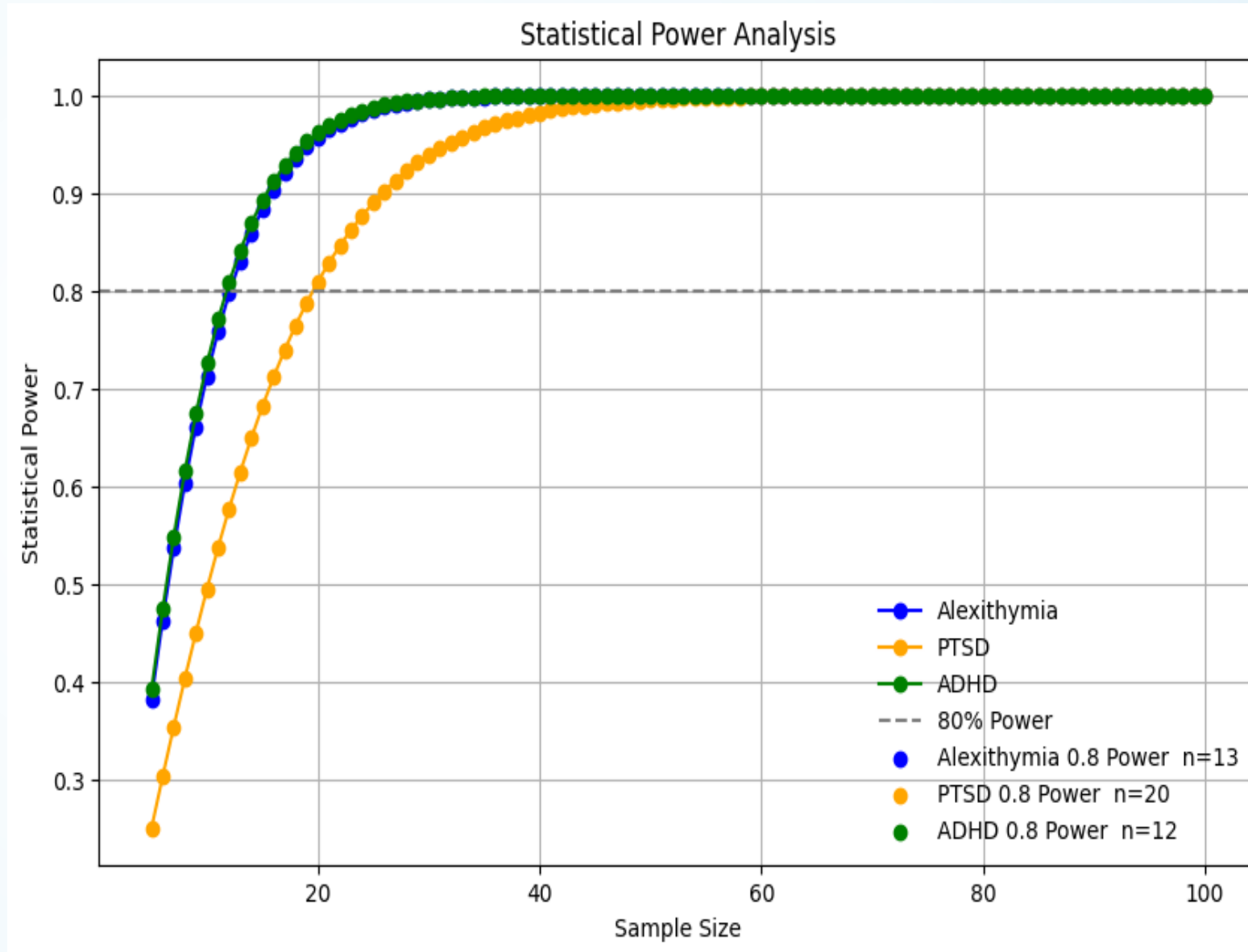


Statistics tests – Power Analysis

$$1 - \beta = \Phi \left(\frac{(d \cdot \sqrt{n})}{\sigma} - z_{\frac{\alpha}{2}} \right)$$

where:

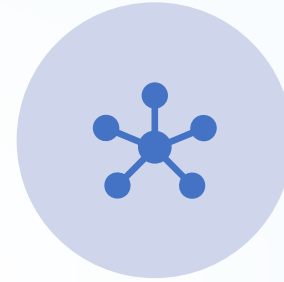
- ✓ $1 - \beta$ is the power of the test.
- ✓ d is the effect size (Cohen's d).
- ✓ n is the sample size.
- ✓ σ is the standard deviation.
- ✓ $z_{(\frac{\alpha}{2})}$ is the critical z-value.
- ✓ Φ is the standard normal CDF.



Conclusions



LLMs can effectively assess mental conditions.



Direct method is more accurate

Considers the broader context and uses its generalized intelligence.



Reliance on self-reporting is problematic.



Individual item predictions leads to error accumulation

Future Work



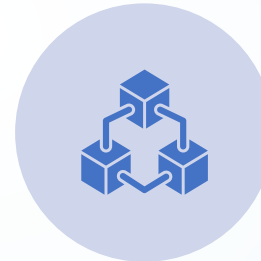
Analyze the visual context (video) of the interview



Benchmark different LLMs



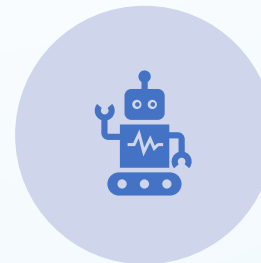
Expand the dataset



Hybrid Approach:
Combine direct and sub-score methods



Fine tune the LLM and the prompt design



Deploy a specialized bot



THANK YOU!